

Experiment 7: Functional Prompt Development – Design Specialized Prompts for Summarization, Email Creation, and Content Generation

Aim

To implement functional prompts using the OpenAI API for summarization, professional email creation, and content generation.

Python Code:

```
from openai import OpenAI

# Paste your OpenAI API Key here
client = OpenAI(
    api_key="sk-proj-FboLgW768fa2IYU2-KXDkmHy_cK9JgnW_P9A-A_3SStE-
Fm94OYnHlDpB6mo5qbWfba9c7LnGdT3BlbkFJoqosff_LYE2VyPCdbNGX_4p3ghTZhcd
Bt4Ff_Fh0tYqmNnhjB64OmSAx8KtGx7X9XtRDiN0cUA"
)

# ----- Summarization Prompt -----
summarization_prompt = """
Summarize the following paragraph in 50 words.

Artificial Intelligence is transforming healthcare by improving disease diagnosis,
supporting robotic surgery, enabling personalized treatment, and enhancing patient
monitoring. AI helps doctors analyze medical data quickly, resulting in better
decision-making and improved patient care.
"""

response = client.responses.create(
    model="gpt-5.4-mini",
    input=summarization_prompt
)

print("===== SUMMARIZATION OUTPUT =====")
print(response.output_text)

# ----- Email Creation Prompt -----
email_prompt = """
Write a professional email requesting one day of leave due to illness.
"""

response = client.responses.create(
    model="gpt-5.4-mini",
    input=email_prompt
```

```
)

print("\n===== EMAIL OUTPUT =====")
print(response.output_text)

# ----- Content Generation Prompt -----
content_prompt = """
Write a promotional social media post for an AI Workshop.
"""

response = client.responses.create(
    model="gpt-5.4-mini",
    input=content_prompt
)

print("\n===== CONTENT GENERATION OUTPUT =====")
print(response.output_text)
```

Output

===== SUMMARIZATION OUTPUT =====

AI is transforming healthcare by improving diagnosis, assisting robotic surgery, enabling personalized treatments, and enhancing patient monitoring. It helps doctors analyze medical data faster, leading to smarter decisions, better efficiency, and improved patient care overall.

===== EMAIL OUTPUT =====

Subject: Leave Request for One Day Due to Illness

Dear [Manager's Name],

I am writing to inform you that I am unwell and am unable to attend work today. I kindly request one day of leave due to illness.

I will keep you updated on my condition and hope to return tomorrow if I feel better. Please let me know if you need any additional information.

Thank you for your understanding.

Sincerely,

[Your Name]

===== CONTENT GENERATION OUTPUT =====

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Result

The functional prompts were successfully implemented using the OpenAI API for summarization, email creation, and content generation. The generated outputs were accurate, well-structured, and relevant to their respective tasks, demonstrating the effectiveness of functional prompt development.